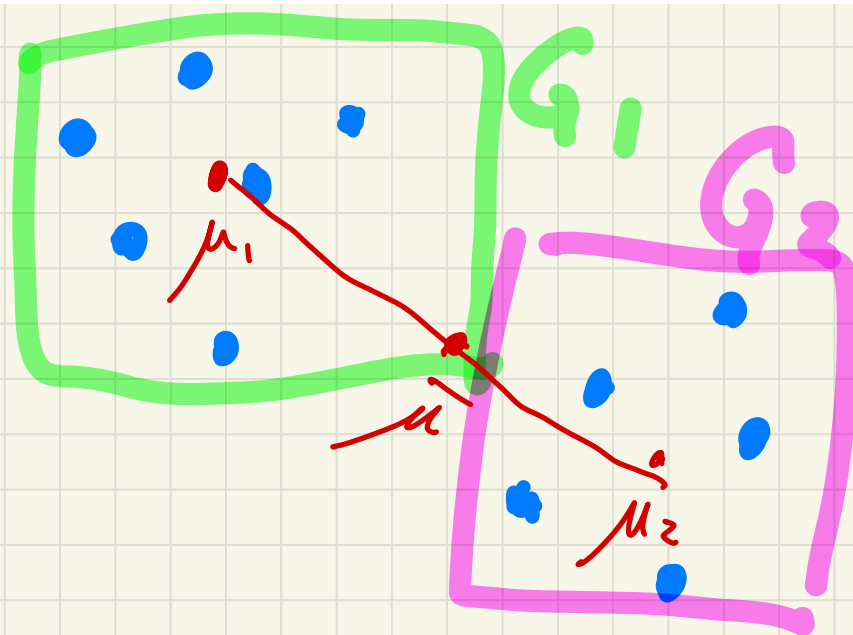


判別分析

2つの母集団 G_1, G_2

勉強時間 $\dots x$



仮定 正規分布 $N(\mu_1, \sigma^2)$ $N(\mu_2, \sigma^2)$

$$\underline{f_1(x)} = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu_1)^2}{2\sigma^2}\right\} = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{D_1^2}{2}\right)$$

x の確率密度関数 $x \in G_1$ $D_1^2 = \frac{(x-\mu_1)^2}{\sigma^2}$

$$x \in G_2 \quad f_2(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{D_2^2}{2}\right) \quad D_2^2 = \frac{(x - \mu_2)^2}{\sigma^2}$$

D_1^2, D_2^2 : マハラノビス距離の2乗
 (サンプルの測定値から
 母集団へのフォリ)

判別方式

$$d(x) = \frac{D_2^2 - D_1^2}{2} = \frac{D_2^2(x) - D_1^2(x)}{2}$$

$$\begin{cases} d(x) \geq 0 \quad (D_2^2 \geq D_1^2) \rightarrow x \in G_1 \\ d(x) < 0 \rightarrow x \in G_2 \end{cases}$$

$$d(x) = \frac{D_2^2 - D_1^2}{2} = \frac{\mu_1 - \mu_2}{\sigma^2} (x - \bar{\mu})$$

$$\bar{\mu} = \frac{\mu_1 + \mu_2}{2}$$

$$D_2^2 = \frac{(x - \mu_2)^2}{\sigma^2} = \frac{x^2 - 2\mu_2 x + \mu_2^2}{\sigma^2}$$

$$D_1^2 = \frac{(x - \mu_1)^2}{\sigma^2} = \frac{x^2 - 2\mu_1 x + \mu_1^2}{\sigma^2}$$

$$\begin{aligned} D_2^2 - D_1^2 &= \frac{(x^2 - 2\mu_2 x + \mu_2^2) - (x^2 - 2\mu_1 x + \mu_1^2)}{\sigma^2} \\ &= \frac{2(\mu_1 - \mu_2)x + \mu_2^2 - \mu_1^2}{\sigma^2} \end{aligned}$$

$$\mu_2^2 - \mu_1^2 = (\mu_1 - \mu_2) \times (\mu_1 + \mu_2)$$

$$\frac{D_2^2 - D_1^2}{2} = \frac{\mu_1 - \mu_2}{\sigma^2} x + \frac{\mu_2^2 - \mu_1^2}{2\sigma^2}$$

$$= \frac{\mu_1 - \mu_2}{\sigma^2} \left(x - \frac{\mu_1 + \mu_2}{2} \right)$$

$$= \frac{\mu_1 - \mu_2}{\sigma^2} (x - \bar{\mu}) \quad \bar{\mu} = \frac{\mu_1 + \mu_2}{2}$$

μ_1 、 μ_2 を $\mathbb{R}^n$ から「決定」

← 推定値

$$\hat{\mu}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_{i1} = \bar{x}_1$$

← $G_1 = \{x_{11}, x_{21}, \dots, x_{n_1,1}\}$

集合 (有限集合)

$$n_1 = |G_1|$$

集合の元の数 (濃度)

(= 5)

$$n_2 = |G_2|$$

(= 5)

$$\hat{\mu}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_{i2} = \bar{x}_2$$

$$\hat{\sigma}^2 = \frac{\sum (x_{i1} - \bar{x}_1)^2 + \sum (x_{i2} - \bar{x}_2)^2}{n_1 + n_2 - 2}$$

$$\hat{d}(x) = \frac{\hat{\mu}_1 - \hat{\mu}_2}{\hat{\sigma}_2} \left(x - \hat{\mu} \right) \quad \hat{\mu} = \frac{\hat{\mu}_1 + \hat{\mu}_2}{2}$$

$$\hat{d}(x) \geq 0 \Rightarrow x \in G_1$$

$$\hat{d}(x) < 0 \Rightarrow x \in G_2$$

補正表

G₁

i	x_{i1}	$x_{i1} - \bar{x}_1$	$(x_{i1} - \bar{x}_1)^2$
1	5.2	-0.8	0.64
2	6.4	0.4	0.16
3	6.5	0.5	0.25
4	4.4	-1.6	2.56
5	7.5	1.5	2.25

$\bar{x}_1 = 6.0$

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5.86

$\bar{x}_2$

i	x_{i2}	$x_{i2} - \bar{x}_2$	$(x_{i2} - \bar{x}_2)^2$
6	4.7	0.3	0.09
7	6.2	1.8	3.24
8	2.5	-1.9	3.61
9	3.6	-0.8	0.64
10	5.0	0.6	0.36
Σ	22.0	0	7.94

$\bar{x}_2 = 4.4$

$$\hat{\mu}_1 = 6.0 \quad \hat{\mu}_2 = 4.4$$

$$\hat{\sigma}^2 = \frac{5.86 + 7.94}{5 + 5 - 2} = \frac{13.8}{8} = 1.725$$

$$\hat{\mu} = \frac{\hat{\mu}_1 + \hat{\mu}_2}{2} = \frac{6.0 + 4.4}{2} = \frac{10.4}{2} = 5.2$$

多変量判別分析の推定式

$$\hat{d}(x) = \frac{\hat{\mu}_1 - \hat{\mu}_2}{\hat{\sigma}^2} (x - \hat{\mu})$$

$$= \frac{6.0 - 4.4}{1.725} (x - 5.2) = \frac{1.6}{1.725} (x - 5.2)$$

$$\approx 0.928 (x - 5.2)$$

0.927536...

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Step 3

免状取得時間6日以内の学生 $x = 6$

$$\begin{aligned}\hat{d}(x) &= 0.928(6 - 5.2) - 0.928 \times 0.8 \\ &= 0.9424 \geq 0\end{aligned}$$

この学生は 合格